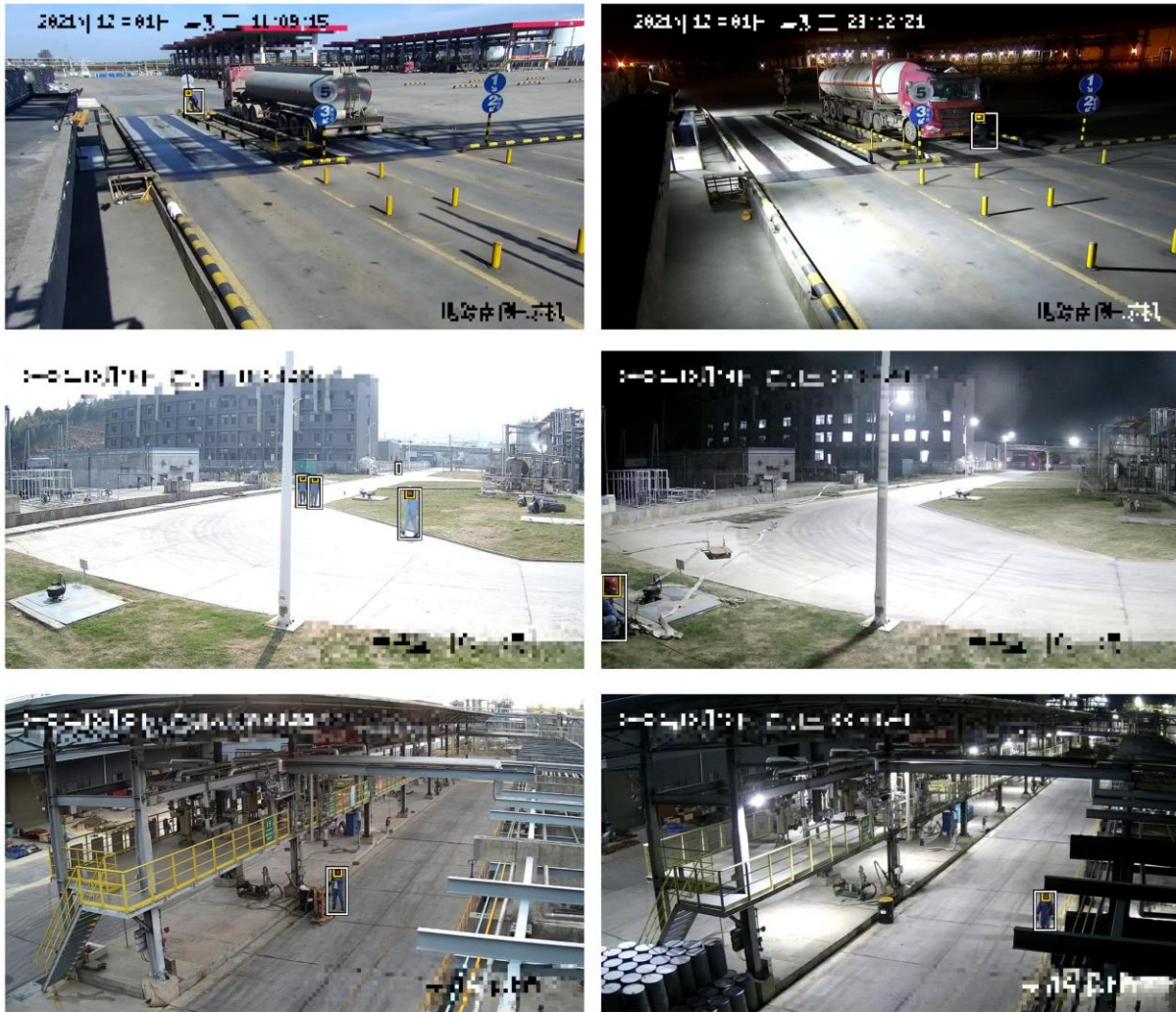




(a) Daytime test frames

(b) Night-time test frames from the same cameras

Figure S1 - Examples of the SFCHD test frames: (a) daytime and (b) night-time frames from the same

three cameras, drawn at random, with their Person (white) and helmet (yellow) labels (Source: Authors'

own work using images from Yu et al. (2023))

Table S1 - Detection performance with a 50-epoch schedule for all detectors and the urban CycleGAN

(mean $\pm$ standard deviation over seeds 0, 1 and 2) (Source: Authors' own work)

Test set	Detector	Precision	Recall	F1-score	mAP@0.5	mAP@0.5:0.95
Construction-PPE test (day, 11 classes)	Baseline Model	0.601 $\pm$ 0.016	0.506 $\pm$ 0.018	0.549 $\pm$ 0.007	0.524 $\pm$ 0.006	0.263 $\pm$ 0.003
	Size-matched control	0.582 $\pm$ 0.012	0.522 $\pm$ 0.025	0.550 $\pm$ 0.010	0.518 $\pm$ 0.013	0.258 $\pm$ 0.003
	Enhanced Model, urban CycleGAN	0.646 $\pm$ 0.055	0.526 $\pm$ 0.010	0.579 $\pm$ 0.017	0.529 $\pm$ 0.003	0.262 $\pm$ 0.001
	Change, control vs Baseline	-3.0%	+3.2%	+0.2%	-1.1%	-1.8%

Test set	Detector	Precision	Recall	F1-score	mAP@0.5	mAP@0.5:0.95
SFCHD day (Person, helmet)	Change, urban Enhanced vs Baseline	+7.6%	+3.9%	+5.4%	+1.0%	-0.2%
	Change, urban Enhanced vs control	+11.0%	+0.7%	+5.2%	+2.1%	+1.6%
	Baseline Model	0.288 ± 0.083	0.169 ± 0.018	0.207 ± 0.010	0.123 ± 0.024	0.027 ± 0.005
	Size-matched control	0.282 ± 0.086	0.158 ± 0.030	0.196 ± 0.016	0.111 ± 0.010	0.024 ± 0.003
	Enhanced Model, urban	0.314 ± 0.039	0.151 ± 0.014	0.203 ± 0.013	0.126 ± 0.008	0.029 ± 0.005
	CycleGAN					
SFCHD night (Person, helmet)	Change, control vs Baseline	-2.0%	-6.2%	-5.7%	-9.8%	-12.3%
	Change, urban Enhanced vs Baseline	+9.1%	-10.4%	-2.1%	+2.7%	+7.7%
	Change, urban Enhanced vs control	+11.3%	-4.5%	+3.8%	+13.9%	+22.9%
	Baseline Model	0.339 ± 0.082	0.168 ± 0.016	0.222 ± 0.022	0.139 ± 0.018	0.032 ± 0.002
	Size-matched control	0.309 ± 0.037	0.148 ± 0.021	0.200 ± 0.026	0.099 ± 0.013	0.021 ± 0.003
	Enhanced Model, urban	0.309 ± 0.050	0.146 ± 0.021	0.196 ± 0.015	0.111 ± 0.013	0.026 ± 0.005
	CycleGAN					
	Change, control vs Baseline	-8.8%	-11.8%	-10.0%	-29.1%	-32.6%
	Change, urban Enhanced vs Baseline	-8.9%	-13.1%	-11.8%	-20.1%	-16.9%
	Change, urban Enhanced vs control	-0.1%	-1.5%	-2.0%	+12.7%	+23.3%

Note: With 50 epochs, the size-matched control and the Enhanced Model (2,264 images) received twice the parameter updates of the Baseline Model (1,132 images). Training used PyTorch 2.10; other settings as in Table 2.

Table S2 - Configuration selection on the night-time validation set (88 MOCS photographs, Person class, 171 instances; one training run per configuration) (Source: Authors' own work)

Detector	CycleGAN	Epochs	Precision	Recall	mAP@0.5	mAP@0.5:0.95	Construction-PPE validation mAP@0.5:0.95
Baseline Model	None	50	0.294	0.158	0.143	0.049	0.283
Size-matched control	None	25	0.381	0.123	0.128	0.045	0.277
Enhanced Model, all translated images	Urban	25	0.555	0.187	0.198	0.061	0.276
Enhanced Model, all translated images (selected)	Construction-night	25	0.540	0.187	0.193	0.065	0.277
Enhanced Model, all translated images	Construction-night	50	0.526	0.187	0.165	0.040	0.274
Enhanced Model, random half of translated images	Construction-night	33	0.425	0.216	0.183	0.055	0.277
Enhanced Model, random half of translated images	Construction-night	50	0.446	0.175	0.141	0.045	0.272

Note: Schedules of 25 epochs (all translated images, 2,264 training images) and 33 epochs (half, 1,698 training images) follow the update-matched rule of Equation 5, rounded to whole epochs. The last column gives the best value on the Construction-PPE validation split, which selects the checkpoint. The construction-night CycleGAN used for this selection was trained with the same settings as the one used for the reported results.

19 Table S3 - Per-seed mAP@0.5 and mAP@0.5:0.95 of the four detectors on the three test sets (Source:
20 Authors' own work)

Test set	Detector	Metric	Seed 0	Seed 1	Seed 2
Construction-PPE test (day, 11 classes)	Baseline Model	mAP@0.5	0.529	0.538	0.532
		mAP@0.5:0.95	0.266	0.265	0.266
	Size-matched control	mAP@0.5	0.533	0.532	0.537
		mAP@0.5:0.95	0.266	0.263	0.270
	Enhanced Model, urban CycleGAN	mAP@0.5	0.531	0.540	0.551
		mAP@0.5:0.95	0.268	0.274	0.277
SFCHD day (Person, helmet)	Enhanced Model, construction-night CycleGAN	mAP@0.5	0.542	0.537	0.542
		mAP@0.5:0.95	0.267	0.273	0.274
	Baseline Model	mAP@0.5	0.130	0.110	0.106
		mAP@0.5:0.95	0.028	0.022	0.024
	Size-matched control	mAP@0.5	0.207	0.138	0.162
		mAP@0.5:0.95	0.048	0.028	0.035
SFCHD night (Person, helmet)	Enhanced Model, urban CycleGAN	mAP@0.5	0.174	0.162	0.190
		mAP@0.5:0.95	0.038	0.036	0.045
	Enhanced Model, construction-night CycleGAN	mAP@0.5	0.176	0.174	0.170
		mAP@0.5:0.95	0.042	0.035	0.035
	Baseline Model	mAP@0.5	0.140	0.120	0.124
		mAP@0.5:0.95	0.030	0.026	0.027
	Size-matched control	mAP@0.5	0.185	0.152	0.144
		mAP@0.5:0.95	0.046	0.034	0.034
	Enhanced Model, urban CycleGAN	mAP@0.5	0.120	0.138	0.151
		mAP@0.5:0.95	0.027	0.028	0.035
	Enhanced Model, construction-night CycleGAN	mAP@0.5	0.174	0.162	0.146
		mAP@0.5:0.95	0.042	0.035	0.034

21 Note: The values are those of the detectors in Table 2, whose means and standard deviations they give.